Generative Adversarial Networks (GANs)

Sanjaya Edirisinghe

Sanjayae@bu.com

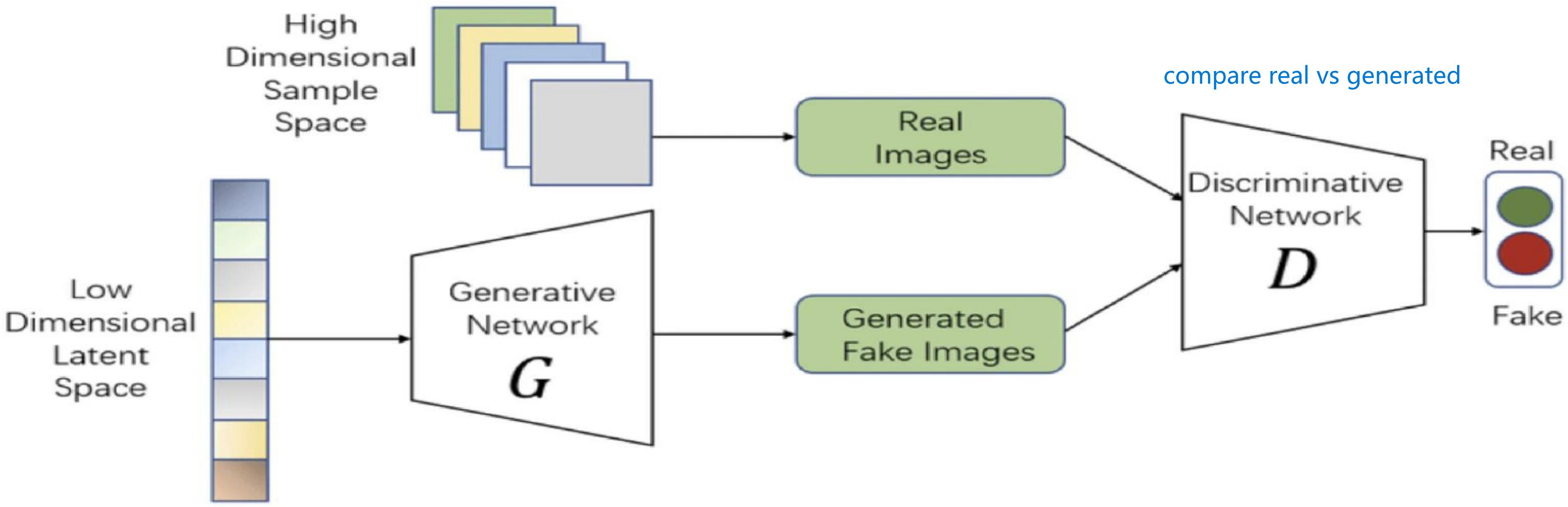
What Are GANs?

- Built on deep **Neural Networks**
- **Invented by Ian Goodfellow** (Google DeepMind team) in 2014
- GANs are **unsupervised learning models** that can generate new, realistic data samples (e.g., images, music, text)

GAN Architecture

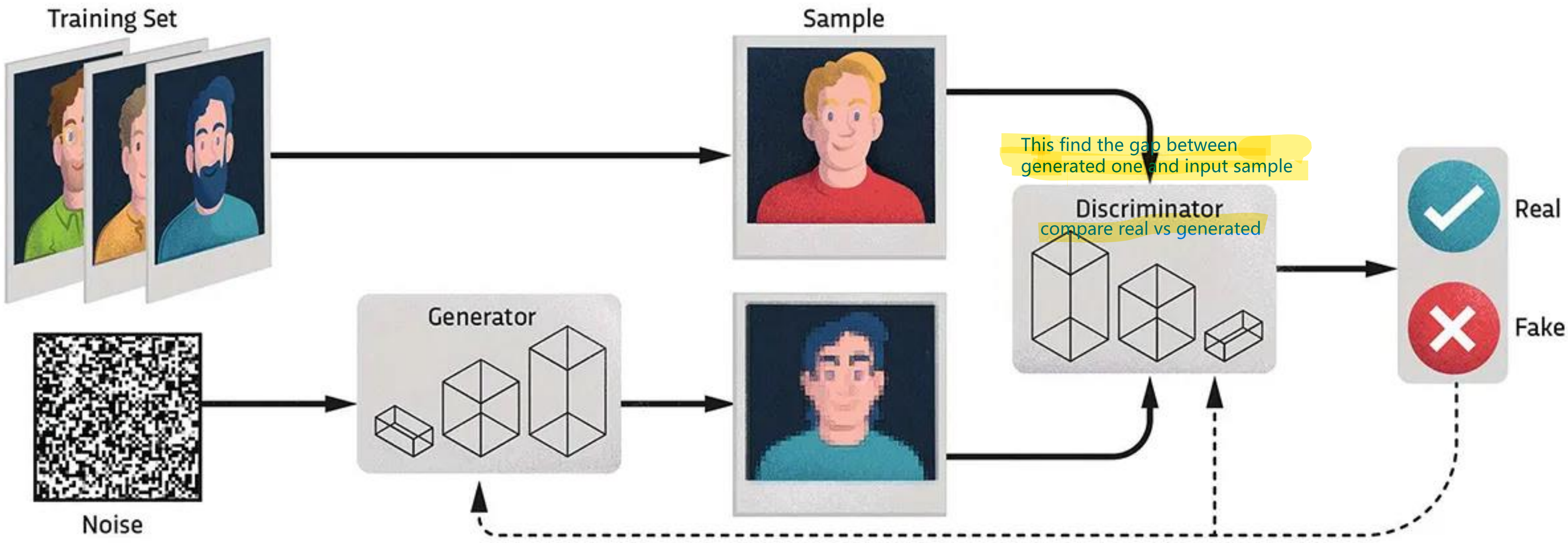
sample of images

compare real vs generated



More clear

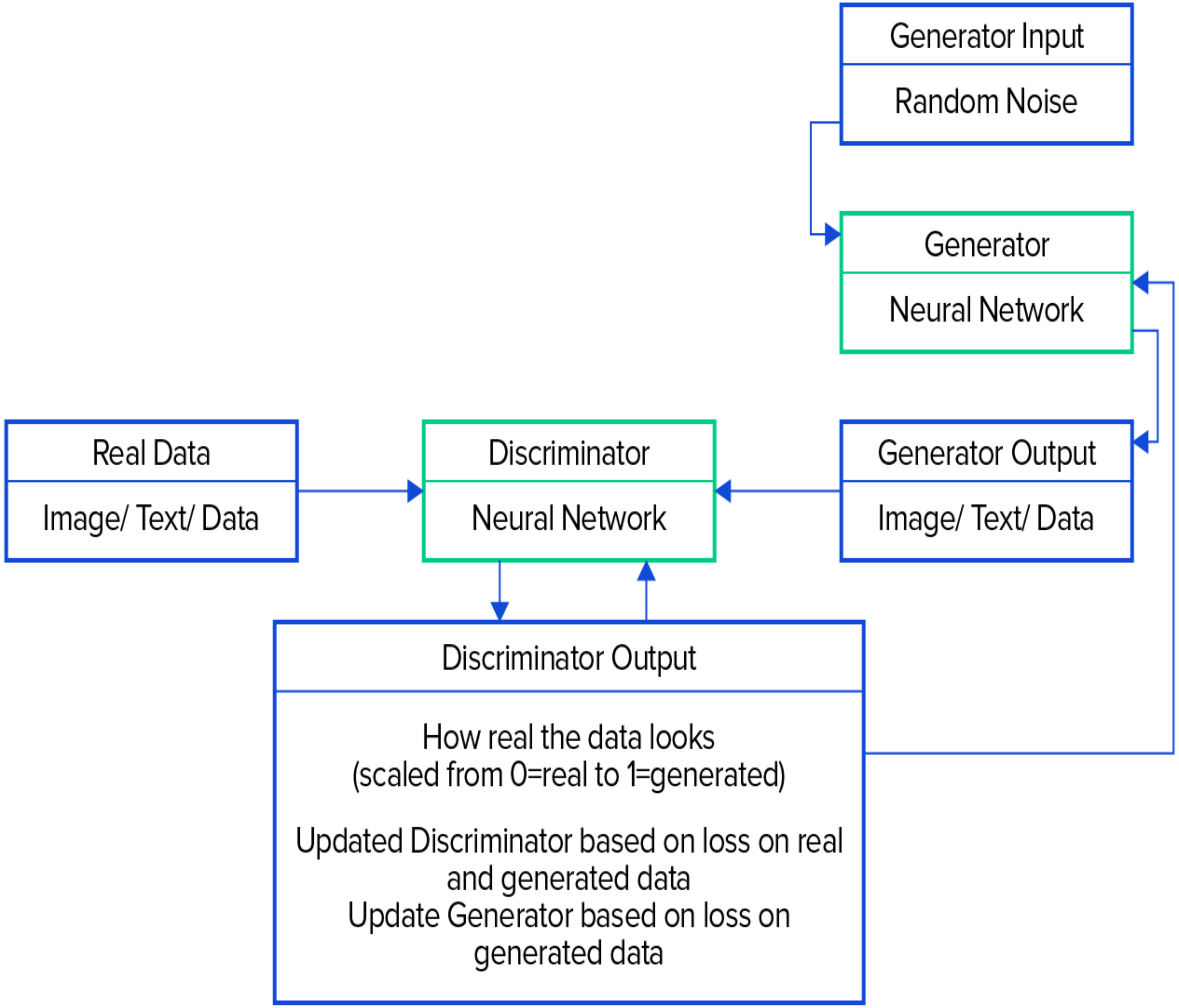
How GANs works ?



Back Propagation

GAN Workflow

Flow diagram of above chart



Adversarial Loss

- **Discriminator Loss:**

$$L_D = -[\log D(x) + \log(1 - D(G(z)))]$$

- **Generator Loss:**

$$L_G = -\log D(G(z))$$

> **Goal:** The generator improves until the discriminator is no better than random guessing (i.e., 50/50).

Training Challenges with GANs

Mode Collapse: Generator produces limited diversity.

Non-convergence: The adversarial training doesn't reach equilibrium.

Vanishing Gradients: Discriminator becomes too strong early on.

Common GAN Variants

Common GAN Variants:
They are different types of Generative Adversarial Networks (GANs) designed for specific tasks like image generation, translation, or improving image quality.

Conditional GAN (cGAN): Incorporate additional information (like labels) to guide the generator in producing specific outputs. For example, a cGAN can be trained to generate images of a particular category (e.g., cats or dogs).

Generates images based on given labels or conditions (e.g., generate a cat or dog).

CycleGAN: Excel at image-to-image translation, particularly when paired data is unavailable. They learn to map between different domains or styles, such as transforming photos into paintings.

Used for image-to-image translation without paired data (e.g., photo → painting).

StyleGAN: Developed by NVIDIA, is known for its ability to produce high-quality, photorealistic images, particularly of portraits.

Generates very realistic high-quality images, especially human faces.

Super-Resolution GAN(SRGAN): focus on enhancing the resolution of low-resolution images, creating high-quality, detailed outputs.

Improves image resolution (low-quality image → high-quality image).

Deep Convolutional GAN (DCGAN): leverage deep convolutional neural networks for generating high-resolution images, making them well-suited for tasks like image synthesis and generation.

Uses CNN layers to generate images, commonly used for image generation tasks.

Conditional GAN (cGAN)

Conditional GANs (cGANs)

Vanilla GANs generate outputs without control.

Conditional GANs allow us to **control the output** by conditioning on a label or input data.

How it works:

Input: Digit label "5"

Output: Generated image that looks like the handwritten digit "5"

Applications:

Text-to-image generation

Image colorization

Supervised image generation

CycleGANs

How it works

Translate images from one domain to another **without paired data**.

Architecture

- Two generators: G (A→B) and F (B→A)
- Two discriminators: One for each domain

Key Idea

Use **cycle consistency**: If we convert image A → B, we should be able to recover A by going B → A

Example

Convert photos of horses to zebras, or summer to winter scenes.

Applications

- Style transfer
- Unpaired image translation (e.g., satellite → map)

StyleGANs

Motivation:

Generate **high-quality, controllable images** with style features (e.g., face generation).

Innovations:

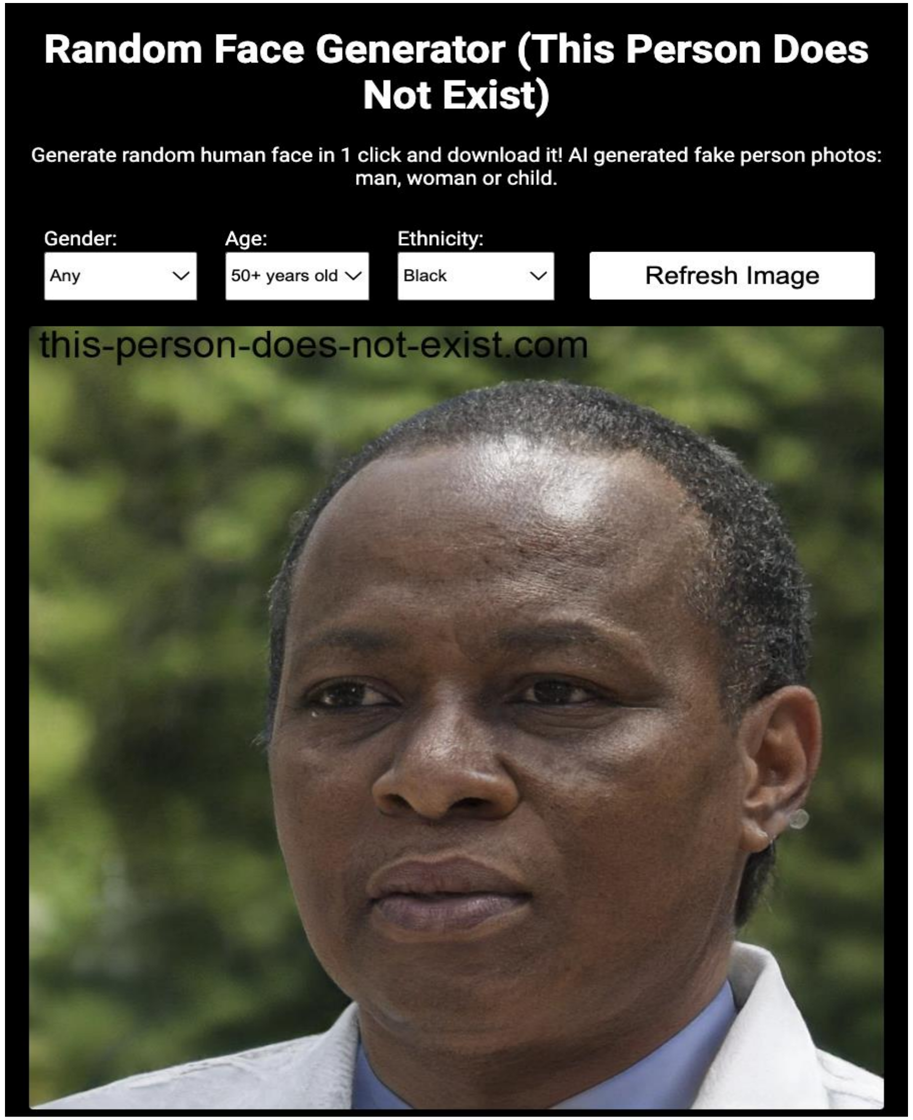
- **Style-based generator**: Controls features at each layer (coarse to fine).
- **Latent space disentanglement**: Enables control over features like pose, expression, age, etc.
- **Progressive growing**: Train from low to high resolution.

Output Example:

- Photorealistic images of faces that **don't exist**

Applications:

- Deepfakes
- Fashion design
- Game character creation.



https://this-person-does-not-exist.com/en